Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/27/2007
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Radiative Enhancement Effects on Flame Spread (REEFS): A Space Flight Experiment Proposal		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	04/27/2003	**End Date:**	01/03/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Gokoglu, Suleyman A	**Contact Phone:**	216-433-5499
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1056		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The research is divided into three topics: 1) radiation-dominated flame spread over thick fuel beds, 2) concurrent-flow flame spread rates and extinction conditions effects of convective and radiative losses, and 3) addition of gaseous fuels to the oxidizing atmosphere. Flame spread over flat solid fuel beds is a useful means of understanding more complex, two-phase, non-premixed flames, such as those found in fires in enclosures, e.g., manned spacecraft and terrestrial buildings. Both transport and chemical effects are well known to affect the characteristics of flame spread. Conductive, convective and radiative transport processes may couple to each other and to chemical effects in unusual ways. Consequently, the goal of this work is to extend previous studies by employing μg experiments with emphasis on the atmospheres and flow		

Task Description:	environments likely to be present in fires which might occur at µg. The goal of this work is to provide information that will enhance crew safety in manned spacecraft including the Space Shuttle, International Space Station, and potential future missions to the moon and Mars. The understanding and control of accidental fires is a critical safety issue in both terrestrial and space-borne environments. Space environments are different from earth-based environments for several reasons. Of course, the major difference is the lack of buoyancy-induced convection. Another difference is that fires in spacecraft may occur in non-standard atmospheres because of the use of non-standard breathing atmospheres or the use of diluents as fire extinguishing agents. Consequently, the understanding of these effects is critical to understanding how fires might start, spread, and be extinguished at µg conditions. One of the tasks of this project, the study of CO2 as a fire extinguishing agent, is particularly relevant to fire safety on the International Space Station, whose fire suppression system uses CO2 as the extinguishing agent. Our experiments show that CO2 is less effective that other agents such as helium at microgravity, whereas the opposite is true at earth gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation is the main mechanism of fire spread between buildings, thus understanding radiation effects on fires is useful to fire prevention and suppression on earth as well as in space.
Task Progress:	The rates of flame spread and minimum oxygen concentrations supporting flame spared over thermally-thick fuel beds were determined at earth gravity and microgravity for varying concentrations of a gaseous extinguishing agent (N2, CO2 or He) added to “baseline” O2-N2 atmospheres. Drop towers were used to obtain microgravity, and foam fuels were employed to obtain sufficiently rapid flame spread in these short-duration facilities. At earth gravity, CO2 was the most effective on a molar basis at reducing spread rate and causing flame extinction whereas at microgravity, He was the most effective. These findings are proposed to result from the transition from a high-speed blowoff-type limit at earth gravity to a low-speed radiative heat loss-induced limit at microgravity. CO2 is less effective as an extinguishment agent at microgravity because it re-absorbs radiation from flame and re-radiates it back to the fuel bed at microgravity, whereas with N2 and He this phenomena is not evident at microgravity because these two gases are radiatively non-participating. He is particularly effective at microgravity because its high thermal diffusivity leads to much larger flame thicknesses resulting in much greater volume for radiative heat losses. Radiative effects are unimportant at earth gravity because convective flow is significantly faster, leading to thinner flames with much lower ratios of radiative to conductive/convective heat transfer. These results are particularly noteworthy considering that the International Space Station employs CO2 fire extinguishers; our results suggest that helium may be a better extinguishing agent on both mass and mole bases at microgravity even though CO2 is much better on a mole basis at earth gravity.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Articles in Peer-reviewed Journals	Son Y, Zouein G, Gokoglu S, Ronney PD. "Comparison of carbon dioxide and helium as fire extinguishing agents for spacecraft." Journal of ASTM International, 2006 Mar;3(3):JAI13564. http://dx.doi.org/10.1520/JAI13564 , Mar-2006